

Stainless steel surfaces with thiol-terminated hyperbranched polymers for functionalization *via* thiol-based chemistry†

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Wen Jing Yang,^a Koon-Gee Neoh,^b En-Tang Kang,^{*b} Serena Lay-Ming Teo^{*c} and Daniel Rittschof^{*d}

Hyperbranched polyethyleneimine (BPEI) was coupled to a polydopamine-coated stainless steel (SS) substrate. Subsequent mercaptoethylation of BPEI with ethylene sulfide produced thiol functional groups on the SS surface. Functionalization of the surface was achieved by end-capping of the hyperbranches with organic molecules *via* thiol-based chemistry, including thiol-epoxy coupling, thiol-ene radical photo-addition and thiol-Michael addition. The SS-P(HEMA-*b*-SBMA) surface was prepared *via* thiol-ene radical photo-addition of the hyperbranches with an alkene-functionalized poly(2-hydroxyethyl methacrylate) (alkene-PHEMA) from atom transfer radical polymerization (ATRP) and subsequent block copolymerization of the zwitterionic monomer, *N*-(3-sulfopropyl)-*N*-(methacryloxyethyl)-*N,N*-dimethyl ammonium betaine (SBMA). The SS-PPEGMA and SS-PMETA surfaces were prepared, respectively, by thiol-initiated photopolymerization of poly(ethylene glycol)methyl ether methacrylate (PEGMA) and 2-(methacryloyloxy)ethyl trimethylammonium chloride (META). The antifouling SS-P(HEMA-*b*-SBMA) and SS-PPEGMA surfaces exhibit resistance to bacterial adhesion, while the SS-PMETA surface is bactericidal. Metal surfaces with thiol-terminated hyperbranches thus provide a versatile platform for tailoring surface functionalities.

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1 Introduction

Surface functionalization of metals plays an important role in applications ranging from biomedical to marine areas.^{1–4} Tethering of polymers onto metal surfaces is an effective approach to functionalize material surfaces.^{5–9} While linear polymer brushes prepared by surface grafting chemistry are useful, grafting of hyperbranched polymers on surfaces has high potential.^{10–14} Hyperbranched polymers are one type of dendritic polymer with a highly branched backbone and a large number of terminal functional groups.¹² They exhibit distinctive chemical and physical properties, such as low intrinsic viscosity, high solubility, good miscibility and poly-functionality.^{11,12,15} Thus, grafting of hyperbranched polymers to substrate surfaces is employed in drug delivery,^{16,17} protein

purification and detection,^{18,19} antibacterial and antifouling substrates^{14,20–23} and cellular engineering.^{24,25}

‘Click’ chemistry, such as copper-assisted azide-alkyne cycloaddition (CuAAC) and strain-promoted azide-alkyne cycloaddition (SPAAC), plays an important role in efficient polymer synthesis and macromolecular engineering.^{26–29} Apart from click chemistry, thiol-based coupling reactions have also been used to construct polymeric structures with high efficiency and specificity.^{30–33} Introduction of thiol functional groups on substrates allows further surface functionalization *via* reactions with electron-rich enes (radical reaction), alkynes (radical reaction), electron-deficient enes (Michael addition), isocyanates (carbonyl addition), epoxies (S_N2 ring opening reaction) and halogens (S_N2 nucleophilic substitution).³¹ Thiol-ene photopolymerization provides a facile approach to the production of surface-grafted polymer films and brushes due to its high polymerization rate and conversion efficiency.^{34,35} The development of a thiol-functionalized platform is of great interest in surface modification and functionalization.

Branched polyethyleneimine (BPEI) is a highly branched polymer containing 25% primary, 50% secondary and 25% tertiary amine groups. The amine groups are chemically reactive and enable a wide variety of chemical modifications. BPEI can thus be employed as a platform for the fabrication of hyperbranched polymeric structures. In the present work, a simple and

^aNUS Graduate School for Integrative Science and Engineering, National University of Singapore, Kent Ridge, Singapore 117576

^bDepartment of Chemical & Biomolecular Engineering, National University of Singapore, Kent Ridge, Singapore, 119260. E-mail: cheket@nus.edu.sg

^cTropical Marine Science Institute, National University of Singapore, Kent Ridge, Singapore, 119223. E-mail: tmsteolm@nus.edu.sg

^dDuke University Marine Laboratory, Nicholas School of the Environment, 135 Duke Marine Lab Road, Beaufort, North Carolina 28516-9721, USA. E-mail: ritt@duke.edu

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versatile strategy for the modification of stainless steel (SS) surfaces with thiol-terminated hyperbranches for thiol-based functionalization was developed *via* mercaptoethylation of BPEI^{36–38} immobilized on the SS surface. Initially, BPEI was coupled to the polydopamine-coated SS surface *via* Schiff-base and Michael addition reactions³⁹ to serve as a hyperbranched platform. Upon mercaptoethylation, the hyperbranched thiol-terminated BPEI on the surface was further functionalized by end-capping with organic molecules in thiol-ene radical photo-addition, thiol-epoxy reaction and thiol-Michael addition. The hyperbranched SS-P(HEMA-*b*-SBMA) surface was prepared by end-capping the thiol-terminated BPEI on SS surfaces with alkene-functionalized poly(2-hydroxyethyl methacrylate) (alkene-PHEMA) *via* thiol-ene radical photo-addition and subsequent atom transfer radical polymerization of zwitterionic monomer *N*-(3-sulfopropyl)-*N*-(methacryloxyethyl)-*N,N*-dimethyl ammonium betaine (SBMA). The hyperbranched SS-PPEGMA surface and the SS-PMETA surface were synthesized, respectively, by terminal polymerization of poly(ethylene glycol)methyl ether methacrylate (PEGMA) and 2-(methacryloyloxy)ethyl trimethylammonium chloride (META) in thiol-initiated photopolymerization from the thiol-terminated BPEI on SS surfaces. The antimicrobial activity of the polymer-functionalized surfaces was assessed with the Gram-positive *S. epidermidis* bacterial cultures.

2 Materials and methods

2.1 Materials

AISI type 304 stainless steel (SS) foils of 0.05 mm thickness were purchased from Goodfellow Ltd. of Cambridge, UK. Branched polyethyleneimine (BPEI, average $M_n \sim 10\,000$ by gel permeation chromatography), 2-hydroxyethyl methacrylate (HEMA, 97%), poly(ethylene glycol)methyl ether methacrylate (PEGMA, $n \sim 8.5$, average molecular weight $M_n \sim 475$), *N*-(3-sulfopropyl)-*N*-(methacryloxyethyl)-*N,N*-dimethyl ammonium betaine (SBMA, 97%), 2-(methacryloyloxy)ethyl trimethylammonium chloride solution (META, 80 wt% in H_2O), glycidyl 2,2,3,3,4,4,5,5,6,6,7,7,8,8,9,9-hexadecafluorononyl ether (GHE, 96%), 2,2,3,3,4,4,4-heptafluorobutyl acrylate (HFA, 97%), allyl 2-bromo-2-methylpropionate (ABM, 98%), triethylamine (TEA, 99%), 10-undecenyl 2-bromoisobutyrate (95%), ethylene sulfide (ES, 98%), dopamine hydrochloride (98%), hexylamine (99%), 2,2'-bipyridyl (BPY, 99%), copper(I) bromide (CuBr, 99%), tris(2-carboxyethyl)phosphine hydrochloride (98%) and 2,2-dimethoxy-2-phenylacetophenone (DMPA, 99%) were purchased from Sigma-Aldrich Co., St. Louis, MO. HEMA and PEGMA were passed through an inhibitor-removal column (Sigma-Aldrich) and then stored under an argon atmosphere at $-10\text{ }^\circ\text{C}$. *Staphylococcus epidermidis* (*S. epidermidis*, ATCC 35984) was obtained from American Type Culture Collection, Manassas, VA, USA. A BacLight Bacterial Viability Kit L131152 was purchased from Molecular Probes Inc., Eugene, Oregon, USA.

2.2 Alkene-PHEMA synthesis

Alkene-functionalized poly(2-hydroxyethyl methacrylate) (alkene-PHEMA) was synthesized *via* atom transfer radical

polymerization (ATRP) of HEMA, using 10-undecenyl 2-bromoisobutyrate as the ATRP initiator to introduce an alkene end group in the polymer chain. In a typical polymerization reaction, 10-undecenyl 2-bromoisobutyrate (24 μl , 8.2×10^{-2} mmol), BPY (25.6 mg, 0.16 mmol) and HEMA (2 ml, 16.4 mmol) were dissolved in 3 ml of methanol. CuBr catalyst (12 mg, 8.2×10^{-2} mmol) was added after the stirred mixture had been degassed for 30 min. After 4 hours, the dark-brown reaction solution turned blue upon exposure to air. The blue mixture was purified by passing it through a silica column to remove the copper catalyst yielding a colorless solution which was dialyzed in ethanol for 3 days and then precipitated into diethyl ether to produce a white solid. ^1H NMR (400 MHz, $\text{DMSO-}d_6$): δ 4.8 (1H); 3.9 (2H); 3.6 (2H); 1.8 (2H); 0.7–1.0 (3H) (Fig. S1, ESI[†]). Gel permeation chromatography (GPC): $M_n = 41\,600\text{ g mol}^{-1}$, polydispersity index (PDI) = 1.19 (Fig. S2, ESI[†]).

2.3 Immobilization of hyperbranched polyethyleneimine on the stainless steel (SS) surface

The polydopamine-coated stainless steel (SS-PDA) substrates were prepared following procedures reported in the literature.³⁹ The hyperbranched polyethyleneimine-functionalized stainless steel substrates (SS-BPEI) were prepared by Schiff-base reaction and Michael addition of BPEI with the SS-PDA surface as follows: the SS-PDA substrate was immersed in the aqueous BPEI solution (0.1 g ml^{-1}) with stirring at room temperature for 12 h; the resulting SS-BPEI substrate was washed thoroughly with doubly distilled water and then stored in a vacuum desiccator after drying under reduced pressure.

2.4 Preparation of a hyperbranched thiol-functionalized SS surface

To prepare a hyperbranched thiol-functionalized SS surface (hyperbranched SS-SH), an SS-BPEI substrate was introduced into 5 ml of ethanol in a reaction flask. The solvent was degassed with argon for 15 min and 25 μl of ethylene sulfide (ES, 0.42 mmol) was added to the solution drop-wise using a syringe under an argon atmosphere. The reaction mixture was degassed for 15 min and the flask was sealed. The reaction was carried out at room temperature for 0.5 h. The substrate was then washed thoroughly with ethanol and doubly distilled water. For comparison, a linear thiol-functionalized SS surface (SS-SH-linear) was prepared by the silanization of the SS surface with 3-mercaptopropyl trimethoxysilane.⁴⁰ The SS substrate was introduced into a 20 mM ethanol solution of 3-mercaptopropyl trimethoxysilane for 6 h. The silanized SS substrate was transferred into 0.1 M hydrochloric acid (HCl) and hydrolysed for 12 h. The resulting substrate was rinsed with copious amounts of ethanol and doubly distilled water, and then cured at $120\text{ }^\circ\text{C}$ for 2 h.

2.5 Functionalization of the hyperbranched SS-SH surface *via* thiol-ene radical photo-addition and thiol-nucleophile reactions

The hyperbranched SS-SH surface was functionalized with organic molecules. Allyl 2-bromo-2-methylpropionate (ABM) was coupled onto the SS-SH surface *via* thiol-ene radical photo-addition.

The hyperbranched SS-SH substrate was introduced into a Pyrex glass tube containing 0.16 ml of ABM (1.0 mmol), 5 mg of DMPA (0.02 mmol) and 2 ml of ethanol. The reaction mixture was degassed for 30 min with argon and then exposed to UV irradiation for 1 h in a Riko model RH 400-10 W rotary photochemical reactor (manufactured by Riko Denki Kogyo of Chiba, Japan). The resulting SS substrate was denoted as SS-SH-ABM. Thiol–nucleophile reactions, including thiol–epoxy and thiol–Michael addition reactions, were employed to further functionalize the SS-SH surfaces. Briefly, the hyperbranched SS-SH substrate was introduced into 2 ml of ethanol solution containing 50 μ l of TEA (0.35 mmol). Then 0.2 ml of glycidyl 2,2,3,3,4,4,5,5,6,6,7,7,8,8,9,9-hexadecafluorononyl ether (GHE, 0.70 mmol) was added dropwise and the reaction was allowed to proceed at room temperature for 12 h. The resulting SS substrate was denoted as SS-SH-GHE. The hyperbranched 2,2,3,3,4,4,4-heptafluorobutyl acrylate (HFA)-functionalized surface (SS-SH-HFA) was prepared *via* the thiol–Michael addition reaction as follows. The SS-SH substrate was introduced into 1 ml of HFA (5.6 mmol), followed by the addition of 20 μ l of hexylamine (0.15 mmol). The reaction was allowed to proceed at room temperature for 5 h.

2.6 P(HEMA-*b*-SBMA)-functionalized SS surface prepared by thiol–ene photo-addition and atom transfer radical polymerization (ATRP)

Alkene-PHEMA was coupled onto the thiol-functionalized substrate to generate a hydrophilic surface. The hyperbranched SS-SH (or SS-SH-linear) substrate, alkene-PHEMA (4 mg), DMPA (5 mg, 0.02 mmol) and ethanol (2 ml) were introduced into a Pyrex glass tube. A purified argon stream was introduced for 30 min to degas the reaction mixture. The reaction tube was then sealed and subjected to UV irradiation for 1 h. The substrate was washed thoroughly with copious amounts of ethanol and doubly distilled water, and denoted as the hyperbranched SS-PHEMA (or SS-PHEMA-linear) surface. For the subsequent atom transfer radical polymerization of zwitterionic monomer SBMA, the hyperbranched SS-PHEMA (or SS-PHEMA-linear) substrate was introduced into a mixture of isopropanol (IPA) and doubly distilled water (IPA : water = 3 : 2, v/v), containing 1.0 g of SBMA (3.6 mmol) and 11.3 mg of BPY (7.2×10^{-2} mmol). The mixture was stirred and degassed with argon for 30 min prior to the addition of CuBr catalyst (5.2 mg, 3.6×10^{-2} mmol). The polymerization was allowed to proceed at room temperature for 12 h to generate P(HEMA-*b*-SBMA)-grafted surface (hyperbranched SS-P(HEMA-*b*-SBMA) or SS-P(HEMA-*b*-SBMA)-linear).

2.7 PPEGMA- and PMETA-functionalized SS surfaces prepared by thiol-initiated photopolymerization

To prepare PPEGMA-functionalized surface, the hyperbranched SS-SH (or SS-SH-linear) substrate was placed in 2 ml ethanol solution of PEGMA (0.5 ml, 1.1 mmol) in a Pyrex glass tube. The mixture was degassed with argon for 30 min and subjected to UV irradiation at room temperature for 1 hour. The antibacterial PMETA-functionalized surface was prepared *via* thiol-initiated photopolymerization of META (0.25 ml, 1.1 mmol) in 2 ml of aqueous solution. The resulting hyperbranched SS-PPEGMA and

SS-PMETA (or SS-PPEGMA-linear and SS-PMETA-linear) substrates were washed thoroughly with ethanol and doubly distilled water.

2.8 Surface characterization

The chemical composition of polymer-functionalized substrates was determined by X-ray photoelectron spectroscopy (XPS). XPS measurements were carried out on a Kratos AXIS Ultra^{DLD} spectrometer with a monochromatized Al K α X-ray source (1486.6 eV photons), at a constant dwelling time of 100 ms and pass energy of 40 eV. The core-level signals were obtained at a photoelectron take-off angle (α , with respect to the sample surface) of 90°. All binding energies (BEs) were referenced to the neutral C 1s hydrocarbon peak at 284.6 eV. The topographies of the polymer-modified surfaces were studied on glass substrate by atomic force microscope (AFM), using a Nanoscope IIIa AFM from Digital Instrument Inc. The root mean square (RMS) roughness (R_q) was calculated from the roughness profile. The thicknesses of the hyperbranched polymer coatings on glass substrate were determined by AFM scratch analysis.^{41,42} Static water contact angles (CA) of the functionalized SS surfaces were measured at 25 °C, using the sessile drop method with a 2 μ l water droplet, in a telescopic goniometer (model 100-00-(230), Rame-Hart, Inc., Mountain Lake, NJ, USA).

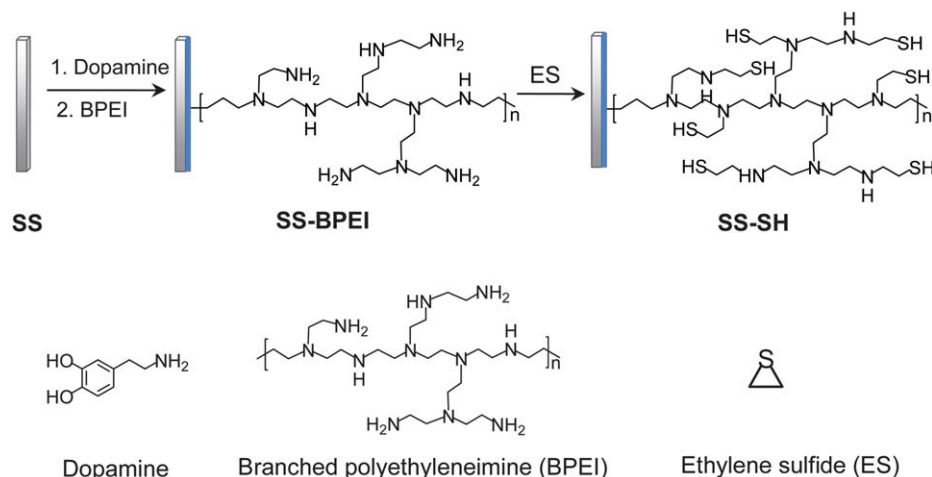
2.9 Bacterial fouling assays

For the comparison, the hyperbranched and linear polymer-functionalized SS surfaces were both tested against Gram-positive *S. epidermidis* (ATCC 35984). The bacteria were cultured in tryptic soy broth at 37 °C overnight. After incubation, the bacterial suspension was centrifuged at 2700 rpm. Upon removal of the supernatant, the bacterial cells were washed with PBS twice and resuspended in a centrifuge tube at a concentration of 5×10^7 cells per ml. Each substrate was immersed in 1 ml bacterial suspension under static condition at 37 °C for 4 h. After mixing with 3% glutaraldehyde and dehydrating with serial ethanol, the adhered bacterial cells were investigated under a scanning electron microscope (SEM, model JSM-5600, JOEL Co., Tokyo, Japan). The live/dead two-color fluorescence method was used to assess the bactericidal effects of polymer-functionalized SS substrates.⁴³ By staining the bacterial cells with a LIVE/DEAD BacLight Bacterial Viability Kit, the viable (appearing green) and dead (appearing red) bacterial cells can be distinguished under a fluorescent microscope. Quantification of bacterial adhesion and viability on the pristine and polymer-functionalized SS was carried out by the spread plate method.⁴³ The result was expressed as a viable adherent fraction, which is defined as the percentage of viable adherent bacteria cells on the polymer-functionalized SS surface relative to those on the pristine SS surface.

3 Results and discussion

3.1 Stainless steel surfaces with thiol-terminated hyperbranches

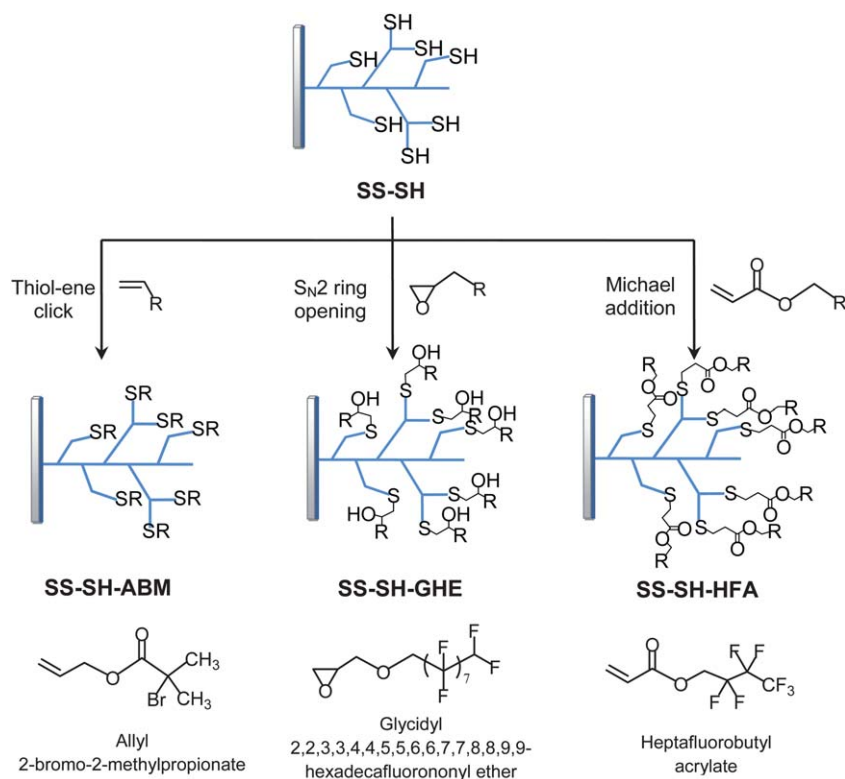
The procedures for preparing stainless steel (SS) surfaces with thiol-terminated hyperbranches are shown in Scheme 1.



Scheme 1 Schematic illustration of the preparation of the hyperbranched SS-SH surface via mercaptoethylation of the polyethyleimine-modified SS surface.

Polydopamine was first coated on the SS surface³⁹ to serve as a bioinspired anchor for coupling of branched polyethyleimine (BPEI) *via* Schiff-base and Michael addition reactions (SS-BPEI surface). The coupled BPEI provided a hyperbranched platform with a large amount of primary, secondary and tertiary amine groups. Ethylene sulfide reacts readily with the primary and secondary amine in BPEI to introduce the thiol functional groups on the surface (SS-SH surface). While direct reaction of ethylene sulfide with the polydopamine-coated surface will also introduce thiol groups, mercaptoethylation of surface-coupled

BPEI can give rise to a hyperbranched thiol-terminated platform for the subsequent construction of hyperbranched surface functionalities. Subsequent functionalization of the surface with thiol-modified hyperbranches (SS-SH surface) was achieved by end-capping with organic molecules *via* thiol-based chemistry, including thiol-epoxy reaction, thiol-ene radical photo-addition and thiol-Michael addition (Scheme 2). The hyperbranched poly(HEMA-*b*-SBMA)-functionalized SS surface was prepared *via* thiol-ene radical photo-addition of alkene-functionalized poly(2-hydroxyethyl methacrylate)



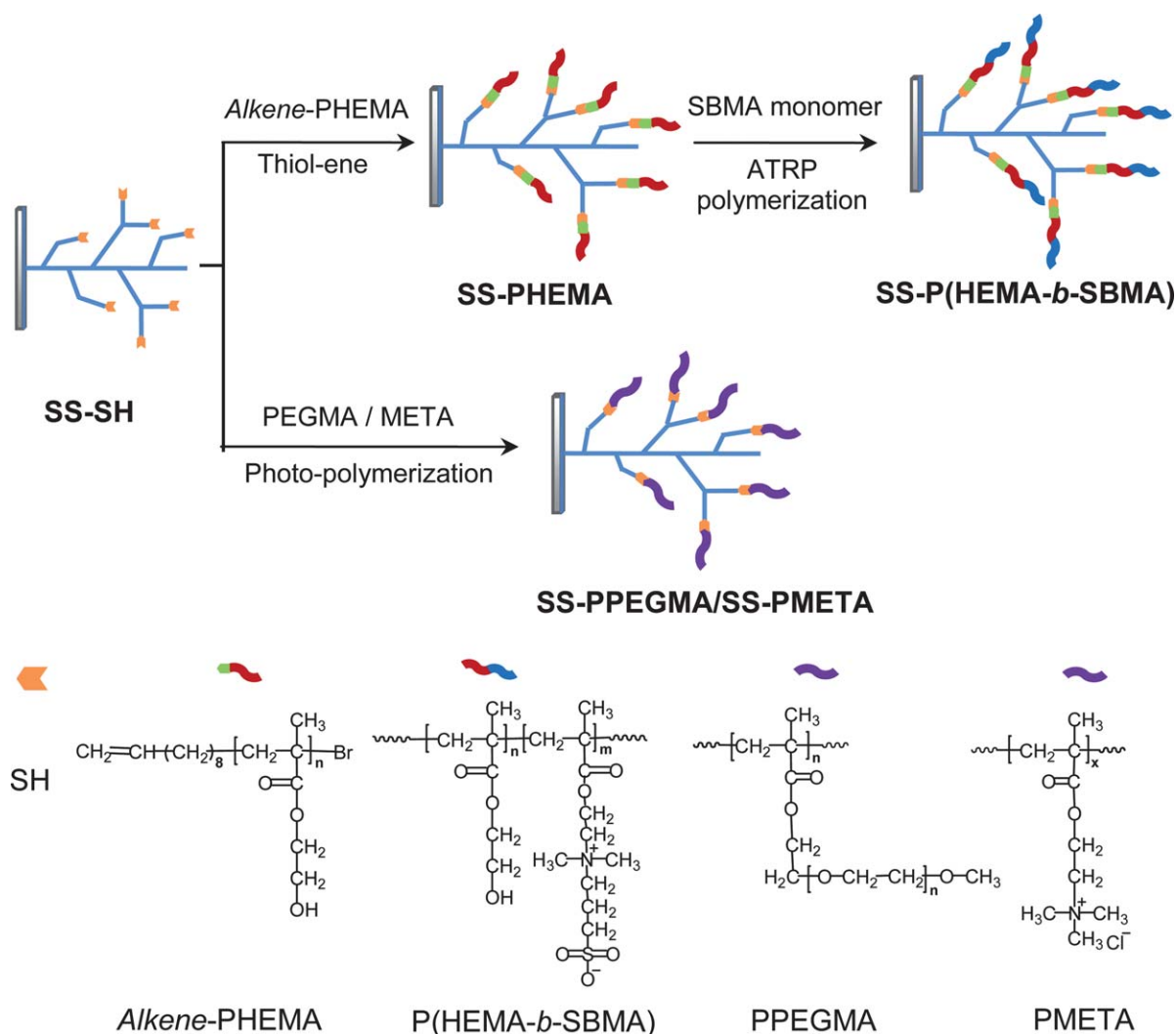
Scheme 2 Schematic illustration of functionalization of the hyperbranched SS-SH SS surface via thiol-based chemistry.

(alkene-PHEMA, prepared *a priori* via atom transfer radical polymerization (ATRP)) and subsequent ATRP of zwitterionic monomer *N*-(3-sulfopropyl)-*N*-(methacryloxyethyl)-*N,N*-dimethyl ammonium betaine (SBMA) (Scheme 3). On the other hand, hyperbranched SS-PPEGMA and SS-PMETA surfaces were prepared *via* thiol-initiated photopolymerization of poly(ethylene glycol)methyl ether methacrylate (PEGMA) and 2-(methacryloyloxy)ethyl trimethylammonium chloride (META), respectively (Scheme 3).

Table 1 shows the static water contact angles of the functionalized SS surfaces. The static water contact angle of the polyethyleneimine-functionalized SS surface (SS-BPEI) is $45 \pm 3^\circ$. The chemical composition of SS-PDA and SS-BPEI surfaces was analyzed by X-ray photoelectron spectroscopy (XPS) (Fig. 1a–d). The XPS C 1s core-level spectra of the SS-PDA (Fig. 1b) and SS-BPEI (Fig. 1d) surfaces are curve-fitted into five peak components with BEs at 284.6, 285.6, 286.2, 287.4 and 288.8 eV, attributable to the C–H, C–N, C–O, C=O and O–C=O species, respectively.^{43,44} The intensity of the C–N peak component has increased substantially after coupling of BPEI on the

SS-PDA surface due to the higher content of C–N species in BPEI. Concomitantly, the [N]/[C] molar ratio, as determined from the XPS N 1s and C 1s core-level spectral area ratio, has increased substantially from 10.9% to 24.4%, consistent with a higher [N]/[C] ratio for BPEI (50%) than that for dopamine (12.5%).

After mercaptoethylation of SS-BPEI with ethylene sulfide, the water contact angle of the thiol-functionalized surface (SS-SH) increases to $53 \pm 3^\circ$. Two additional XPS signals with BEs at 228 and 165 eV, attributable to the S 2s and S 2p species, respectively,⁴⁴ appear in the XPS wide scan spectrum of the SS-SH surface (Fig. 1e). The XPS S 2p core-level spectrum (Fig. 1f) consists of a spin–orbit split doublet, with BEs of the S 2p_{3/2} and S 2p_{1/2} peak components at 163.1 and 164.3 eV, respectively, attributable to the covalently bonded sulfur (C–S) species.^{44,45} The [S]/[N] molar ratio is about 23.7%, as determined from the XPS S 2p and N 1s core-level spectral area ratio, indicating that ethylene sulfide has reacted with about 32% of primary and secondary amine groups in BPEI in the mercaptoethylation process. The apparent surface coverage by thiol groups was



Scheme 3 Schematic illustration of functionalization of the hyperbranched SS-SH surface *via* thiol–ene photo-addition and thiol-initiated photopolymerization.

Table 1 Static water contact angles and surface compositions of the polymer-functionalized stainless steel surfaces

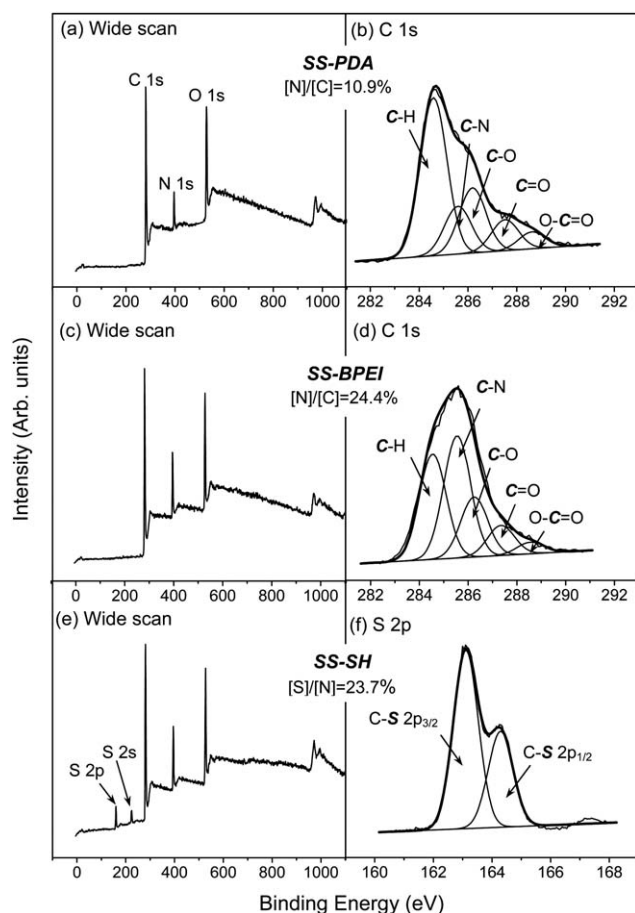
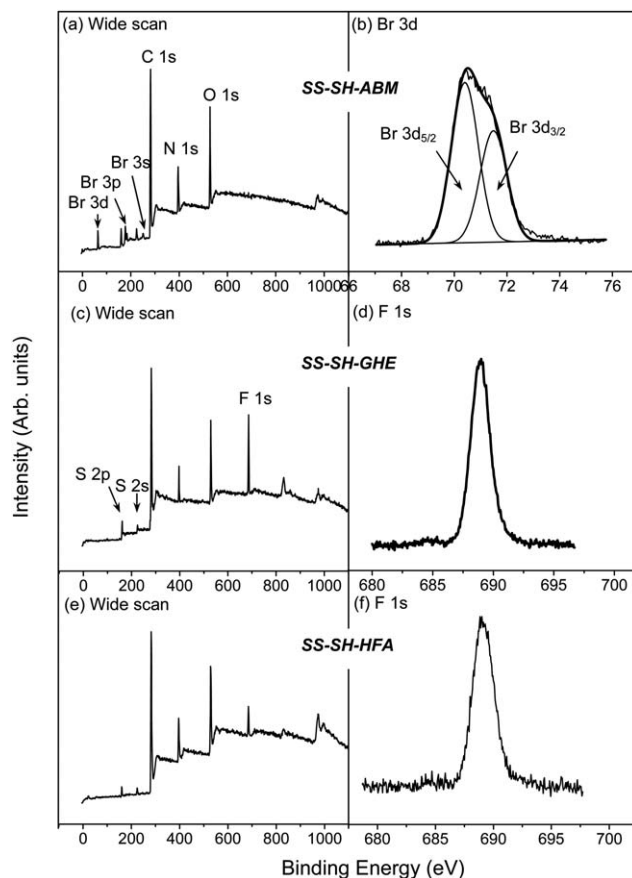
Surfaces	Static water contact angle (mean $\pm$ SD ^a , degrees)	Surface composition (molar ratio)
SS-BPEI	45 $\pm$ 3	[C] : [N] = 4.1 : 1
SS-SH	53 $\pm$ 3	[C] : [O] : [N] : [S] = 23.5 : 5.3 : 4.2 : 1
SS-SH-ABM	57 $\pm$ 3	[S] : [Br] = 1.2 : 1
SS-SH-GHE	91 $\pm$ 3	[F] : [S] = 9.3 : 1
SS-SH-HFA	81 $\pm$ 3	[F] : [S] = 4.5 : 1
SS-PHEMA	45 $\pm$ 3	[S] : [Br] = 1.3 : 1
SS-P(HEMA- <i>b</i> -SBMA)	25 $\pm$ 3	[N] : [S] = 1.4 : 1
SS-PPEGMA	29 $\pm$ 2	[C] : [O] = 3.0 : 1
SS-PMETA	19 $\pm$ 2	[C] : [N] : [Cl] = 22.7 : 1.7 : 1

^a SD denotes standard deviation.

around $10.5 \pm 1.5 \text{ nmol cm}^{-2}$, as determined by iodometric titration (ESI[†]).^{35,46} Therefore, a hyperbranched platform has been created for the subsequent functionalization and modification of the SS surfaces.

3.2 Functionalization of the hyperbranched SS-SH surface via thiol-ene radical photo-addition and thiol-nucleophile reactions

The hyperbranched thiol-functionalized SS surface provides a versatile platform for tailoring surface functionalities. A multitude of organic molecules and polymers can be coupled to the thiol groups on the SS-SH surfaces *via* radical and catalyzed processes under mild conditions.³¹ Allyl 2-bromo-2-methylpropionate (ABM) was successfully coupled to the SS-SH surface *via* the thiol-ene radical photo-addition. The resulting surface composition was revealed by the XPS results. Three XPS peaks with BEs at 70, 189 and 256 eV appear in the wide scan spectrum of the SS-SH-ABM surface (Fig. 2a), associated with the Br 3d, Br 3p and Br 3s core-level signals, respectively.⁴⁴ The corresponding XPS Br 3d core-level spectrum (Fig. 2b) is curve-fitted into two peak components with respective BEs at 70.4 and 71.5 eV for the Br 3d_{5/2} and Br 3d_{3/2} doublet, consistent with the presence of covalently bonded alkyl bromide species on the surface.^{43,44} The coupling efficiency of ABM to the SS-SH surface is about 83%, as determined from the [S]/[Br] molar ratio derived from XPS S 2p and Br 3d core-level spectral area ratio (Table 1). Reaction efficiencies (surface coverage) of 45–75% had been reported for various functional thiols reacting with alkene-terminated monolayers on silicon surfaces under similar

**Fig. 1** XPS wide scan and C 1s core-level spectra of the (a and b) SS-PDA and (c and d) SS-BPEI surfaces; (e and f) XPS wide scan and S 2p core-level spectra of the SS-SH surface.**Fig. 2** (a and b) XPS wide scan and Br 3d core-level spectra of the SS-SH-ABM surface; XPS wide scan and F 1s core-level spectra of (c and d) SS-SH-GHE and (e and f) SS-SH-HFA surfaces.

experimental conditions.³³ The alkyl bromide with an α -carbonyl group can serve as an effective ATRP initiator for subsequent surface-initiated polymerization.⁴⁷

Apart from the thiol-ene photo-addition reaction, thiol-nucleophile reactions, such as the thiol-epoxy ring-opening reaction and thiol-Michael addition, were performed under mild conditions. The thiol-epoxy reaction has been utilized in many important biosyntheses and biomedical processes.³¹ Glycidyl 2,2,3,3,4,4,5,5,6,6,7,7,8,8,9,9-hexadecafluorononyl ether (GHE) was coupled onto the SS-SH surface to achieve a hydrophobic SS-SH-GHE surface. The contact angle of the SS-SH-GHE surface increases (from $53 \pm 3^\circ$ for the SS-SH surface) to $91 \pm 3^\circ$, indicating the hydrophobic nature of the surface (Table 1). Simultaneously, a strong F 1s core-level signal at a BE of 689 eV appears in the XPS wide scan spectrum of the SS-SH-GHE surface (Fig. 2c).⁴⁴ Based on the XPS F 1s and S 2p core-level spectral area ratio, the [F]/[S] molar ratio is determined to be about 9.3 : 1, indicating that about 58% of the thiol groups have reacted with GHE in the thiol-epoxy ring-opening reaction (Table 1). Heptafluorobutyl acrylate (HFA) was coupled onto the hyperbranched SS-SH surface *via* the thiol-Michael addition reaction catalyzed by hexylamine. The successful coupling of HFA was confirmed by the appearance of an F 1s signal with BE at 689 eV in the XPS wide scan spectrum of the SS-SH-HFA surface (Fig. 2e). The water contact angle of the surface was also increased to $81 \pm 3^\circ$ (Table 1). The reaction efficiency of HFA with the SS-SH surface is about 64%, as determined from the surface [F]/[S] molar ratio (Table 1). The modest reaction efficiencies are probably due to steric hindrance and the heterogeneous nature of the reaction on the SS surface.⁴⁸

3.3 Antifouling SS-P(HEMA-*b*-SBMA) surface prepared by thiol-ene radical photo-addition and atom transfer radical polymerization

Microorganisms, such as bacteria, tend to adhere strongly to the surface of materials, followed by rapid biofilm development, leading to failure of the materials.^{2,6,49} Alkene-PHEMA chains were coupled onto the hyperbranched thiol-functionalized surface for antifouling purposes. The preserved dormant alkyl bromide end group of the alkene-PHEMA was then employed to initiate the subsequent ATRP of zwitterionic monomer SBMA.

The hyperbranched SS-SH surface becomes hydrophilic after coupling of alkene-PHEMA and subsequent ATRP of SBMA. The respective static water contact angles of the PHEMA-functionalized surface (SS-PHEMA) and P(HEMA-*b*-SBMA)-grafted surface (SS-P(HEMA-*b*-SBMA)) are $45 \pm 3^\circ$ and $25 \pm 2^\circ$, respectively (Table 1), indicating the hydrophilic nature of surfaces. Fig. 3 shows the XPS wide scan, N 1s and S 2p core-level spectra of the SS-PHEMA and SS-P(HEMA-*b*-SBMA) surfaces. The appearance of three Br signals at the BEs of 70, 189, and 256 eV in the wide scan spectrum of the SS-PHEMA surface (Fig. 3a) indicates the presence of alkyl bromide on the surface⁴⁴ and thus confirms the successful coupling of alkene-PHEMA *via* thiol-ene radical photo-addition. The XPS Br 3d core-level spectrum (inset of Fig. 3a) is curve-fitted into two peak components with BEs at 70.4 and 71.5 eV, attributable to the Br

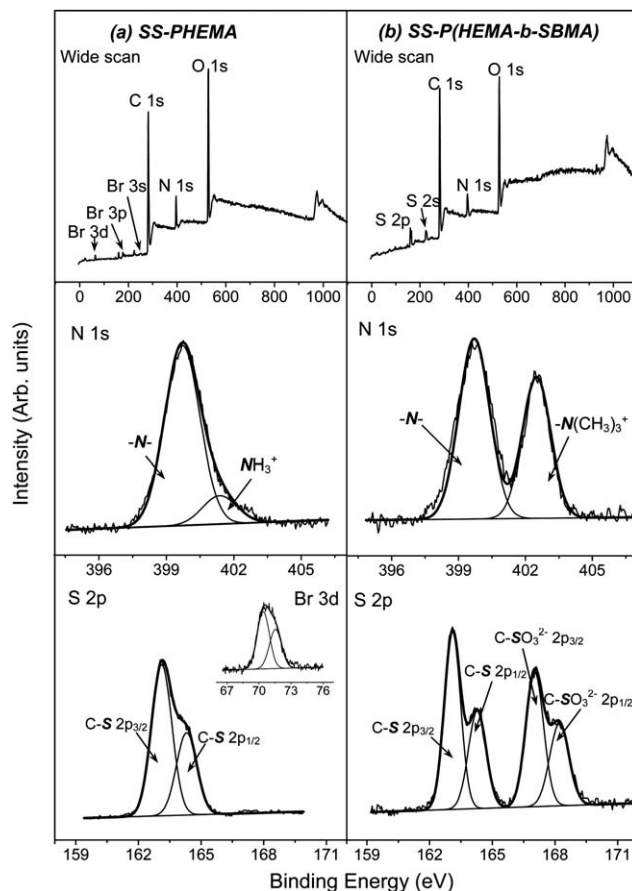


Fig. 3 XPS wide scan, N 1s and S 2p core-level spectra of the hyperbranched (a) SS-PHEMA and (b) SS-P(HEMA-*b*-SBMA) surfaces.

$3d_{5/2}$ and Br $3d_{3/2}$ species, respectively, of covalently bonded bromide species.^{43,44} The XPS S 2p core-level spectrum of the SS-PHEMA surface consists of a spin-orbit split doublet with BEs at 163.1 and 164.2 eV for the S $2p_{3/2}$ and S $2p_{1/2}$ peak components, respectively, of the covalently bonded sulfur (C-S) species.⁴⁴ About 77% of thiol groups have reacted with alkene-PHEMA in thiol-ene photo-addition, as determined from the XPS-derived [S]/[Br] molar ratio (Table 1).

In comparison to the XPS N 1s core-level spectrum of the SS-PHEMA surface (Fig. 3a), a new nitrogen peak component with BE at 402.5 eV appears in the XPS N 1s core-level spectrum of the SS-P(HEMA-*b*-SBMA) surface (Fig. 3b). This peak component is associated with the quaternary ammonium cations ($-N(CH_3)_3^+$) in the side chains of PSBMA.⁴⁴ Concomitantly, two spin-orbit split doublets are discernible in the S 2p core-level spectrum of the SS-P(HEMA-*b*-SBMA) surface (Fig. 3b). The main S 2p doublet with BEs at 163.1 and 164.2 eV for the respective S $2p_{3/2}$ and S $2p_{1/2}$ species is associated with the formation of a normal C-S bond.^{44,45} The other S 2p doublets at the BEs of 167.0 and 168.2 eV are attributed to sulfonate C-SO₃⁻ species^{44,45} in the side chains of PSBMA. The thickness of the P(HEMA-*b*-SBMA) polymer layer was about 53 nm, as determined by AFM depth profiling of the scratched surface (Fig. S3a, ESI†). Thus, the presence of grafted P(HEMA-*b*-SBMA) chains on the surface was confirmed.

3.4 Hyperbranched SS-PPEGMA and SS-PMETA surfaces prepared by thiol-initiated photopolymerization

Thiol-ene photo-polymerization has been applied to the fabrication of a variety of surface-grafted functional polymer brushes.^{34,35,50} The antifouling PPEGMA^{51,52} and antibacterial PMETA^{53,54} brushes were grafted from the respective hyperbranched SS-SH surface *via* thiol-initiated photopolymerization. The SS-PPEGMA surface is hydrophilic with a static water contact angle of $29 \pm 2^\circ$ (Table 1). The N 1s core-level signal at a BE of 399 eV, and S 2p and 2s core-level signals with respective BEs at 165 and 228 eV⁴⁴ have disappeared completely in the wide scan spectrum of the SS-PPEGMA surface (Fig. 4a), suggesting the presence of a dense layer of PPEGMA brush coating on the surface. The thickness of the PPEGMA polymer layer was about 137 nm, as determined by AFM depth profiling of the scratched surface (Fig. S3b, ESI[†]), indicating the high efficiency of thiol-initiated photopolymerization. The XPS C 1s core-level spectrum of SS-PPEGMA surface (Fig. 4b) consists of three peak components with BEs at 284.6, 286.1 and 288.5 eV, attributable to the C-H, C-O and O-C=O species, respectively.^{44,55} The high content of C-O species is consistent with the successful photopolymerization of PEGMA from the surface.

The antibacterial SS-PMETA surface is also hydrophilic in nature, with the static water contact angle of $19 \pm 2^\circ$ (Table 1). In the XPS wide scan spectrum of the SS-PMETA surface (Fig. 4c), the appearance of Cl 2s and Cl 2p signals, with respective BEs at 271 and 197 eV, indicates the successful thiol-initiated photopolymerization of META from the SS-SH surface.⁴⁴ The XPS C 1s core-level spectrum (Fig. 4d) is curve-fitted into four peak components with BEs at 284.6, 285.5, 286.2 and 288.7 eV, attributable to the C-H, C-N, C-O/C-N(CH₃)₃⁺ and O-C=O species.⁴⁴ Moreover, the XPS N 1s core-level spectrum (Fig. 4e) consists of two peak components with BEs at 399.4 and 402.5 eV, associated with the amine (-N-) and quaternary ammonium cations (N(CH₃)₃⁺) in the side chains of PMETA, respectively.⁴⁴ The Cl 2p core-level spectrum with the Cl 2p_{3/2} peak component at a BE of 196.7 eV (Fig. 4f) indicates the presence of Cl⁻ ions as counterions of quaternary ammonium cations (N(CH₃)₃⁺) on the SS-PMETA surface. The PMETA brush layer reached a thickness of about 94 nm (Fig. S3c, ESI[†]) after 1 h of photo-polymerization.

3.5 Morphologies of the hyperbranched polymer-functionalized SS surfaces

The surface morphologies of the polymer-functionalized antimicrobial surfaces are revealed by the AFM images in Fig. 5. The SS-BPEI surface is homogeneous and uniform (Fig. 5a), with a surface root-mean-square (RMS) roughness (R_q) of 0.40 nm. The RMS roughness increases to 0.89 nm after thiolation of ethylene sulfide (Fig. 5b). Coupling of alkene-PHEMA resulted in a rough surface with R_q of 1.96 nm (Fig. 5c). However, the subsequent ATRP of SBMA produced a smooth and dense surface coating (Fig. 5d, $R_q = 1.39$ nm). As for the SS-PPEGMA and SS-PMETA surfaces, prepared by thiol-initiated photopolymerization, dense and uniform polymer brush coatings were grafted from the SS-SH surface, with respective RMS roughness of 1.55 and 0.71 nm (Fig. 5e and f). The thickness of the PPEGMA and PMETA layers are about 137 and 94 nm, respectively, after 1 h of photo-polymerization (Fig. S3b and c, ESI[†]).

To evaluate the stability of polymer brush coatings, the SS-PPEGMA samples were immersed in PBS for a period of one month. The XPS analysis results of the aged SS-PPEGMA surface are shown in Fig. S5a and b (ESI[†]). The lineshapes of the C 1s core-level spectrum of the SS-PPEGMA surface remained practically unchanged after immersion in PBS for one month. The PPEGMA coating thickness remained at around 131 nm after aging/immersion in PBS for one week (Fig. S5c, ESI[†]). The XPS and AFM analysis results thus indicate good stability of the polymer coating in an aqueous environment.

3.6 Bacterial-fouling assays of the hyperbranched polymer-functionalized SS surfaces

The antimicrobial activity of the hyperbranched polymer-functionalized surfaces was evaluated using Gram-positive *Staphylococcus epidermidis* (*S. epidermidis*, ATCC 35984). Fig. 6 shows fluorescence microscopy images of pristine SS and polymer-functionalized SS surfaces after exposure to *S. epidermidis* for 4 h. With live/dead stain, viable cells appear green, while the dead ones appear red under a fluorescence microscope. A high

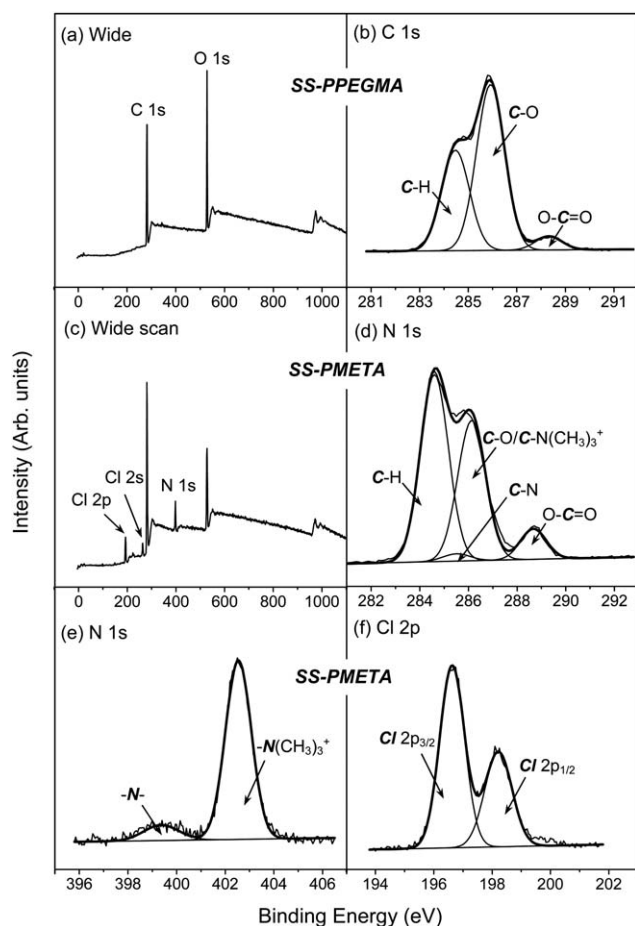


Fig. 4 (a and b) XPS wide scan and C 1s core-level spectra of the hyperbranched SS-PPEGMA surface; (c–f) XPS wide scan, C 1s, N 1s and Cl 2p core-level spectra of the hyperbranched SS-PMETA surface.

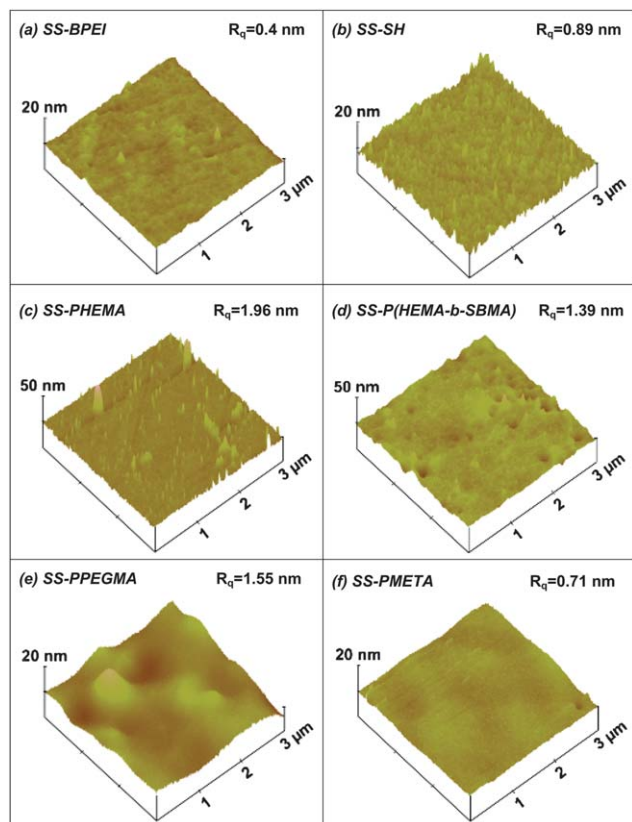


Fig. 5 Atomic force microscopy (AFM) images of the hyperbranched (a) SS-BPEI, (b) SS-SH, (c) SS-PHEMA, (d) SS-P(HEMA-*b*-SBMA), (e) SS-PPEGMA and (f) SS-PMETA surfaces. R_q = root-mean-square roughness.

concentration of viable *S. epidermidis* cells (stained green) were on the pristine SS surface (Fig. 6a), with very few dead cells (stained red) (inset of Fig. 6a), indicating susceptibility of the pristine SS to bacterial fouling. Similar results were observed on the SS-SH surface (Fig. 6b). In comparison to the pristine SS, the amount of viable bacterial cells attached to the hyperbranched SS-PHEMA surface decreased substantially (Fig. 6c), revealing the low-fouling effect of PHEMA brushes. The hyperbranched P(HEMA-*b*-SBMA) coating exhibited even higher resistance to bacterial adhesion, as evidenced by fewer viable bacterial cells on the SS-P(HEMA-*b*-SBMA) surface (Fig. 6d). This result indicates that subsequent polymerization of zwitterionic SBMA monomers, initiated by the dormant bromide groups on the SS-PHEMA surface, improves the resistance to bacterial adhesion. For the hyperbranched SS-PPEGMA surface (Fig. 6e), viable bacteria adhered sparsely on the surface, indicating higher efficacy to anti-bacterial adhesion. The bacterial-resistant efficiencies of the PHEMA, P(HEMA-*b*-SBMA) and PPEGMA brush coatings arise from the high degree of hydration of the polymer brushes in an aqueous environment,³ either induced by hydrogen-bonding or electrostatic interactions.⁵⁶ The presence of a large number of dead cells on the SS-PMETA surface (Fig. 6f) indicated the strong bactericidal action of the PMETA brush coating, which disrupts cell membranes through the actions of quaternary ammonium cations.^{57–59}

In order to compare the antimicrobial efficacies of hyperbranched and linear polymer-functionalized surfaces,

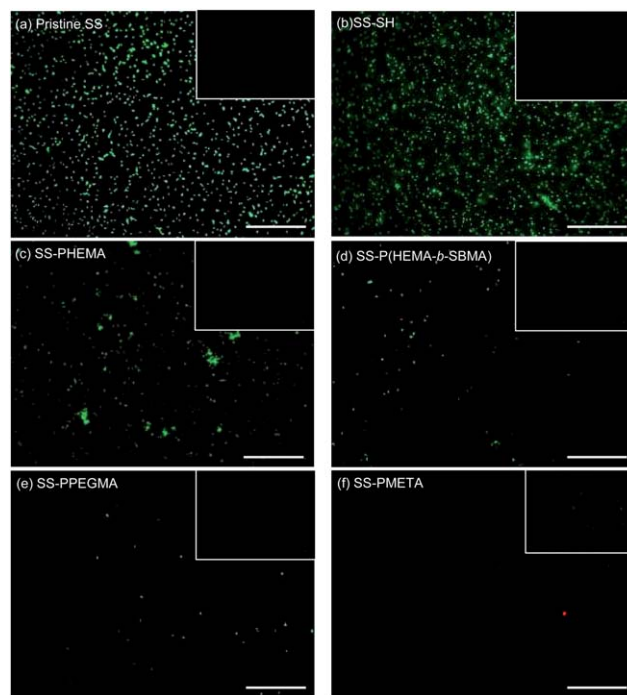


Fig. 6 Fluorescence microscopy images of the (a) pristine SS, and hyperbranched (b) SS-SH, (c) SS-PHEMA, (d) SS-P(HEMA-*b*-SBMA), (e) SS-PPEGMA and (f) SS-PMETA surfaces after exposure to *S. epidermidis* (5×10^7 cells per ml) for 4 h. Scale bar: 50 μ m.

3-mercaptopropyl trimethoxysilane (MPTS) was coupled to the SS surface to provide thiol functional groups for subsequent functionalization. A linear P(HEMA-*b*-SBMA)-functionalized SS surface (SS-P(HEMA-*b*-SBMA)-linear), PPEGMA-functionalized SS surface (SS-PPEGMA-linear) and PMETA-functionalized SS surface (SS-PMETA-linear) were then prepared. Successful syntheses of the linear thiol- and polymer-functionalized SS surfaces were ascertained from the XPS analysis results (Fig. S4, ESI†). Fig. 7 shows the scanning electron microscopy (SEM) images of hyperbranched and linear polymer-functionalized surfaces after bacterial fouling. Bacterial cells adhered readily as individual cells and in clusters on the pristine SS surface (Fig. 7a). A significant reduction in bacterial adhesion was observed on the hyperbranched SS-PHEMA surface (Fig. 7b) as compared to that of the pristine SS surface. In comparison to the SS-PHEMA surface, the number of bacterial cells on the SS-P(HEMA-*b*-SBMA) was further reduced (Fig. 7c), indicating the high resistance of zwitterionic PSBMA to *S. epidermidis*. The SS-P(HEMA-*b*-SBMA)-linear surface exhibited lower efficacy to anti-bacterial adhesion than that of the hyperbranched SS-P(HEMA-*b*-SBMA) surface, as evidenced by the presence of more adherent bacteria cells in Fig. 7d. Only a few bacterial cells adhered to the hyperbranched SS-PPEGMA and SS-PMETA surfaces (Fig. 7e and g), indicative of bacterial resistance and/or bactericidal activity. More bacterial cells were observed on the SS-PPEGMA-linear and SS-PMETA-linear surfaces (Fig. 7f and h) than on the corresponding hyperbranched surfaces, indicating better resistance of the hyperbranched polymer-functionalized surfaces to bacterial fouling.

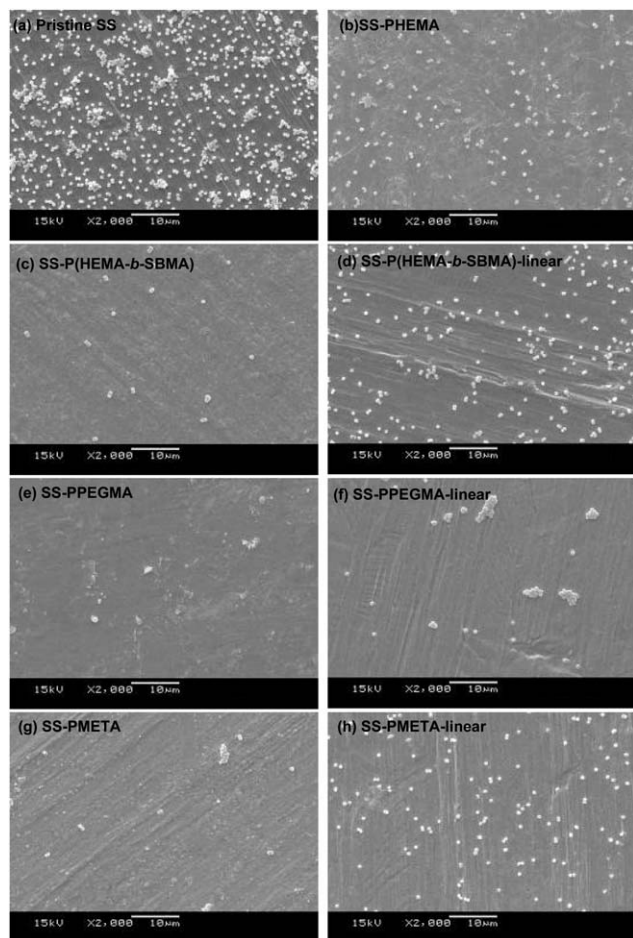


Fig. 7 Scanning electron microscopy (SEM) images of the (a) pristine SS, (b) SS-PHEMA, (c) SS-P(HEMA-*b*-SBMA), (d) SS-P(HEMA-*b*-SBMA)-linear, (e) SS-PPEGMA, (f) SS-PPEGMA-linear, (g) SS-PMETA, (h) SS-PMETA-linear surfaces after exposure to *S. epidermidis* (5×10^7 cells per ml) for 4 h.

A more quantitative *in vitro* antibacterial assay was carried out using the spread plate method. Fig. 8 shows the viable adherent fractions of *S. epidermidis* on the pristine and polymer-functionalized SS surfaces after 4 h of exposure to bacterial suspension. In comparison to the pristine SS surface, all the polymer-functionalized SS surfaces exhibited over 80% reduction in bacterial adhesion. The viable adherent fractions for the linear and hyperbranched polymer-grafted surfaces were 12–18% and 3–10%, respectively. In comparison to the linear polymer-functionalized surface, the hyperbranched polymer-functionalized surface further reduced bacterial adhesion by about 50%, indicating the superior antibacterial efficacy of the hyperbranched polymer coatings. In addition, the hyperbranched SS-PPEGMA and SS-PMETA surfaces exhibit greater reduction in bacterial adhesion than the SS-P(HEMA-*b*-SBMA) surface. This result may be attributed to the denser PPEGMA and PMETA coatings (90–140 nm), prepared by the thiol-initiated photo-polymerization, than the P(HEMA-*b*-SBMA) coating (53 nm), synthesized *via* the thiol-ene photo-addition and ATRP (Fig. S3, ESI†). The high antifouling and antimicrobial efficacies of the hyperbranched polymer-functionalized SS surfaces are thus ascertained by the fluorescence and SEM images, as well as *in vitro* bacterial fouling assays.

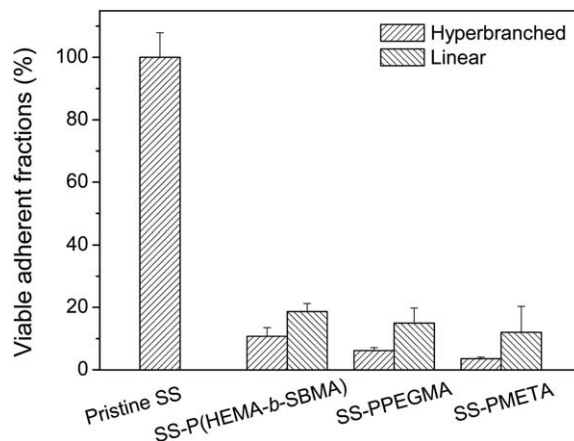


Fig. 8 Viable adherent fractions of *S. epidermidis* on the hyperbranched and linear polymer-functionalized SS substrates after exposure to *S. epidermidis* (5×10^7 cells per ml) for 4 h. The cell number was determined by the spread plate method. Each error bar represents the standard deviation calculated from three replicates.

4 Conclusions

Stainless steel surfaces with thiol-terminated hyperbranched polyethyleneimine were prepared as a platform for functionalization *via* thiol-based chemistry. Successful end-capping of the hyperbranches with organic molecules was *via* thiol-epoxy reaction, thiol-ene radical photo-addition and thiol-Michael addition. Antifouling SS surfaces with hyperbranched P(HEMA-*b*-SBMA) (hydrophilic and zwitterionic block copolymers) were prepared *via* the thiol-ene radical photo-addition and atom transfer radical polymerization. SS surfaces with hyperbranched PPEGMA and hyperbranched PMETA can be readily synthesized by thiol-initiated photopolymerization. In comparison to the linear polymer-functionalized surfaces, the hyperbranched polymer-functionalized SS surfaces exhibit superior efficacies to anti-bacterial adhesion and bactericidal action. Metal surfaces with thiol-terminated polymer hyperbranches thus provide a versatile platform for tailoring the surface functionalities and properties.

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